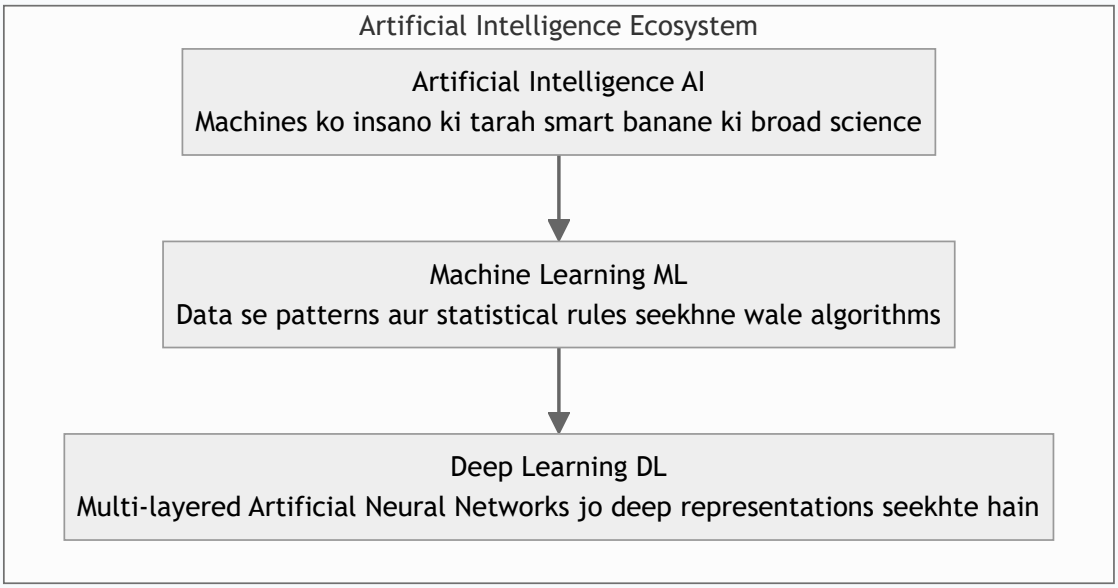
Lecture 36: Deep Learning Foundations & Artificial Neurons (Hinglish)

Swagat hai **Lecture 36: Deep Learning Foundations (Part-1)** ke master curriculum repository mein! Yeh lecture classical Machine Learning se modern Deep Learning aur Artificial Neural Networks ki taraf ek aitihasik transition mark karta hai.

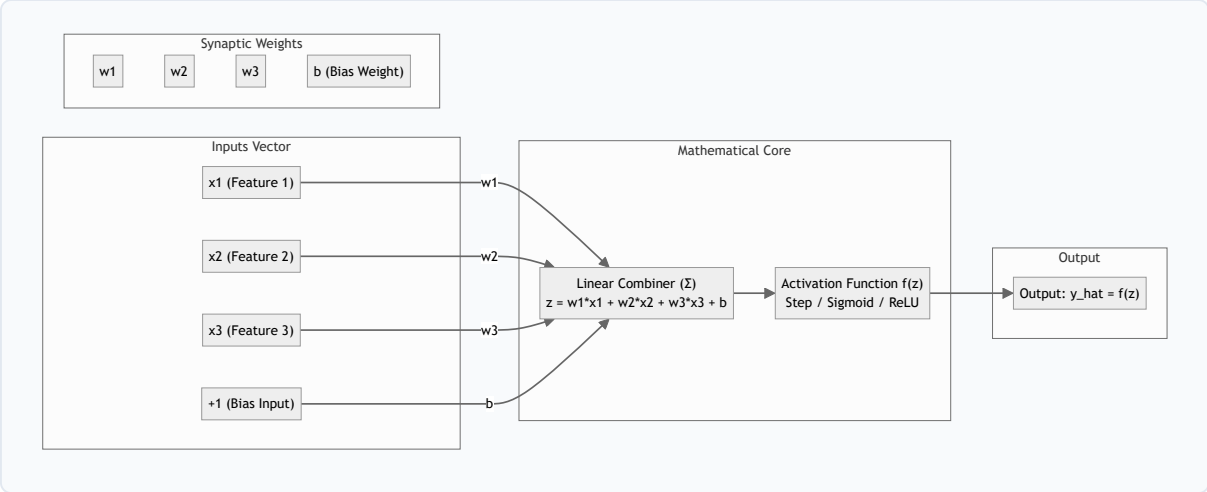
Duniya bhar ki modern AI technologies—jaise self-driving cars ka vision system, ChatGPT aur Claude jaise Large Language Models (LLMs)—multi-layered neural networks ke dum par hi chalti hain. Is module mein humne Perceptron ki mathematics, activation functions, non-linear representation learning, hardware acceleration, aur PyTorch code implementation ko zero se master kiya hai.

AI, ML aur Deep Learning ka Hierarchy



Core Concept: Artificial Neuron (Perceptron)

Neural Network ka sabse basic building block ek artificial neuron (Perceptron) hota hai. Yeh inputs ko weights se multiply karta hai, bias jodta hai, aur activation function se pass karke final output nikalta hai:



Mukhya Mathematical Formula:

$$z = \sum_{i=1}^{n} w_i x_i + b = \mathbf{w}^T \mathbf{x} + b$$

$$\hat{y} = f(z) = f(\mathbf{w}^T \mathbf{x} + b)$$

Comprehensive Submodule Roadmap

Is lecture ko 10 topic submodules aur 1 comprehensive industrial practice suite mein organize kiya gaya hai. Har module mein English aur Hinglish documentation, Mermaid whiteboard diagrams, aur compiled PDFs shamil hain:

Submodule	Topic Title	Key Concepts & Formulas	Asset Reference	Documentation Links
36.01	**What is Deep Learning?**	AI vs ML vs DL hierarchy, Big Data, GPU compute, real-world apps	Diagram / Concepts	English | Hinglish
36.02	**ML vs Deep Learning**	Manual vs Automated feature engineering, Andrew Ng scaling curve	Diagram / Concepts	English | Hinglish

Submodule	Topic Title	Key Concepts & Formulas	Asset Reference	Documentation Links
36.03	**What are Neural Networks?**	Biological neuron mapping, 3-layer ANN architecture (Input, Hidden, Output)	Diagram / Concepts	English | Hinglish
36.04	**Perceptron (In Depth)**	Frank Rosenblatt model, weights, bias necessity, Step, Sigmoid, ReLU, training loop	Diagram / Concepts	English | Hinglish
36.05	**Multi-Layered Networks**	XOR non-linearity limit, Multi-Layer Perceptron (MLP), Universal Approximation	Diagram / Concepts	English | Hinglish
36.06	**PyTorch vs TF vs Keras**	Dynamic vs static computation graphs, research vs enterprise production	Diagram / Concepts	English | Hinglish
36.07	**PyTorch Installation**	CPU vs CUDA GPU installation, hardware detection script	`pytorch_installation.png`	English | Hinglish
36.08	**Google Colab Setup**	Free NVIDIA T4 GPU cloud runtime, Drive integration, session limits	Diagram / Concepts	English | Hinglish

Submodule	Topic Title	Key Concepts & Formulas	Asset Reference	Documentation Links
36.09	**Building a Neuron (Code)**	`torch.tensor`, `nn.Linear(3, 1)`, forward pass, mathematical proof	Notebook	English | Hinglish
36.10	**Neuron Code Walkthrough**	Cell-by-cell notebook execution, `@` matrix dot product verification	Notebook	English | Hinglish
Practice	**Industrial Scenarios**	Cardiovascular diagnostic neuron, XOR mathematical proof, GPU VRAM sizing	Multiple	English Questions | Hinglish Questions | Solutions